

ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

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ABSTRACT

Fundamental questions remain about when and why adversarial examples (AExs) arise across neural networks (NNs), with existing accounts often framing them either as consequences of irregular decision landscapes or as sensitivity to predictive but non-robust input features. We argue that, within a broad class of models that compress many latent features into limited representational dimensions, adversarial vulnerability can also arise from *efficient* information encoding itself. In particular, we show that superposition – the representation of more features than available dimensions – induces latent feature arrangements whose interference structure can be exploited by adversarial perturbations. Across controlled synthetic tasks, CIFAR-10 image models with engineered bottlenecks, and an algorithmic modular-arithmetic setting, attack patterns are predictable from the relevant representation geometry rather than appearing arbitrary. This perspective provides a mechanistic explanation for phenomena such as attack transferability between similarly trained models and class-specific vulnerability patterns, while clarifying the assumptions under which the explanation applies. In synthetic settings with precisely controlled superposition, we establish that superposition *suffices* to create adversarial vulnerability, and we show that the same qualitative signatures persist in a vision transformer (ViT) trained on CIFAR-10. Our results suggest that adversarial vulnerability can be a byproduct of representational compression and feature interference, not only a flaw in learning or a property of non-robust inputs. We release our implementation through a software repository.

1 INTRODUCTION

Despite extensive research on AExs (Szegedy et al., 2014; Goodfellow et al., 2014), there remains no consensus on the mechanisms that make small perturbations effective across different models, tasks, and data domains. We still lack a sufficiently general account of how minimal input changes can substantially alter predictions, limiting our ability to predict successful perturbations, explain transfer between models, or design defences that target the relevant failure mode.

Existing explanations broadly fit into two camps (Nakkiran, 2019). The *bug* perspective attributes vulnerability to factors such as learning flaws (Schmidt et al., 2018; Nakkiran, 2019) or geometric properties of high-dimensional decision boundaries (Fawzi et al., 2016). This line of work often provides statistical bounds on properties such as capacity (Bubeck & Sellke, 2021) or Lipschitz constraints (Hein & Andriushchenko, 2017), but does not usually specify which perturbation directions should succeed in a particular representation. The *feature* perspective argues that AExs exploit predictive but non-robust statistical patterns in data (Ilyas et al., 2019), shifting attention to the relationship between data structure and human perception. However, this view can treat non-robust features as fixed properties of the input distribution, rather than as properties that may also depend on how a model encodes the distribution. Neither approach fully accounts for the interaction between representational constraints, data semantics, and the attack directions available in a trained network.

In this paper, we connect these perspectives using a mechanistic interpretability lens, focusing on how adversarial vulnerability can arise when architectural constraints meet data semantics (*i.e.* human interpretable properties of the data). We show that AExs can emerge from interference between learned representations in NNs. In the settings we study, adversarial attacks systematically exploit superposed latent structure – a mechanism that ordinarily expands representational capacity – to produce perturbations that manipulate model outputs. Our analysis identifies a pathway that can apply wherever similar representational pressures occur: input correlations constrain feature

arrangements, these arrangements determine interference patterns, and the interference patterns shape attack characteristics and transferability. This framework makes the success and transfer of perturbations predictable from learned geometry under the paper’s stated assumptions.

Our account of AExs builds on recent theories of how networks encode and process information, especially the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH posits that *input features*—abstractions of data that may range from simple attributes to semantic concepts—are represented as linear directions in a network’s representation space. Superposition further hypothesises that NNs can represent substantially more such features than they have dimensions, efficiently packing multiple features into shared directions while introducing interference. Because this interference allows perturbing one represented feature to affect others, it provides a candidate mechanism for adversarial vulnerability across settings where feature compression is present. This paper tests that mechanism and uses it to interpret several adversarial phenomena. Our primary contributions are:

- We present a framework that relates architectural constraints, feature properties, and adversarial vulnerability. In simple controlled models, we show how data properties induce superposition geometries that determine realised adversarial perturbations, and how the resulting interference patterns explain attack transferability and class-specific susceptibility.
- We show that the same qualitative signatures persist beyond the toy setting in a ViT trained on CIFAR-10 with an engineered bottleneck layer that induces class-feature superposition.
- We show that superposition is sufficient, but not necessary, for adversarial vulnerability. This distinction identifies algorithmic brittleness as a separate mechanism that can arise even with orthogonal feature representations.
- We demonstrate that mechanistic understanding of learned representations can guide adversarial attack construction, supporting the case for semantically informed defences that target the exploited representation rather than only the input norm.

2 BACKGROUND

We now introduce the representational assumptions used throughout our analysis: the LRH and superposition. Formally, let $\mathbf{x} \in \mathcal{X}$ denote the input to a NN and $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ the activation vector of the l -th layer.

Linear representation hypothesis (LRH). The LRH posits that NNs often represent variables used in their computation, including semantic properties of inputs, as linear directions in activation space that can serve as abstractions for analysis (Park et al., 2024; Guerner et al., 2023). This tendency toward linear feature directions is plausible across many architectures because linear separability makes features easy to recognise and manipulate, while dot products with later-layer weights provide an efficient way to process directional features. Growing empirical work supports this view (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (*e.g.* concepts like “presence of shape,” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions $\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that differ mainly in the value of feature c_j while holding other features approximately constant (*i.e.*, $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k \mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs can represent more latent features than there are available activation dimensions: $M > d_l$. For example, LLMs can refer to many more place names than they

have residual stream dimensions. Superposition provides a general framework for analysing this regime: networks use an overcomplete and non-orthogonal set of feature directions $\{\mathbf{v}_j\}_{j=1}^M$ rather than assigning every feature a separate dimension. This is possible because $2^{\Theta(d\epsilon^2)}$ almost orthogonal vectors ($< \epsilon$ cosine similarity) can be represented in d -dimensional space (Tkocz, 2012), and sparse vectors can be accurately recovered after projection into lower-dimensional spaces (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This compression creates two broadly relevant challenges. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) arises when individual neurons contribute to multiple features, so a neuron’s activation need not correspond to a single interpretable concept. Second, non-orthogonal feature directions create *interference* between features – activating one feature partially activates others. Networks may mitigate this ambiguity through non-linear operations (e.g. ReLU, softmax) that help disambiguate superposed features (Gurnee et al., 2023).

Definition 2 (Superposition Hypothesis) *A network layer represents features in superposition if:*

1. *The number of latent features exceeds layer dimensionality: $M > d_l$*
2. *There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$*
3. *Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$*

The network trades representational capacity against feature interference by packing more features than dimensions.

Adversarial attacks. We use the standard notion of adversarial perturbations and attacks following (Szegedy et al., 2014; Costa et al., 2024):

Definition 3 (Adversarial attack) *Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p -sense, e.g. $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small δ that p -perturbs $\mathbf{x}$ to create an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ (untargeted) or $f(\mathbf{x}') = y_{\text{target}}$ (targeted) is known to be an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are created via projected gradient descent (PGD) (detailed in Sec. A).*

Definition 4 (Adversarial robustness) *A classifier $g_w(\cdot)$ with parameters w is said to be **robust** at $\mathbf{x}$ with radius $\epsilon > 0$, if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.*

3 SUPERPOSITION GEOMETRY CAN DETERMINE ADVERSARIAL ATTACKS

We begin in a controlled setting where the relevant latent features, their correlations, and the amount of representational compression are all known. This lets us test whether adversarial attacks exploit interference rooted in feature superposition, and whether the same mechanism can account for broader phenomena such as transferability. Specifically, we ask three questions:

1. *Do adversarial perturbations exploit the interference between superposed features?*
2. *Do correlations in the input shape the geometric arrangement of superposed latent features?*
3. *Can shared latent geometry explain why attacks transfer between independently trained models?*

Experimental setup. To investigate these questions, we design a synthetic task that instantiates established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022) while keeping the underlying feature structure exactly observable. The task is intended to: (1) provide an intuitive classification setting for studying AExs; (2) represent class concepts as linear directions as assumed by the LRH; (3) induce controlled superposition between latent features; and (4) preserve a priori knowledge of how inputs correspond to the superposed features, enabling direct tests of adversarial mechanisms.

Specifically, we partition input $\mathbf{x} \in \mathbb{R}^d$ into k equally-sized groups $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, where $\mathbf{x}^{(j)}$ represents the p input features associated with class j . The task identifies which group has the largest sum: $y = \operatorname{argmax}_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$. Intuitively, each $x_i^{(j)}$ contributes evidence for class j , with the p features standing in for distinct class attributes. Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ with sparsity S (probability of setting to zero).

A two-layer network compresses the input into m dimensions through a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (where $m < k < d$), and reconstructs class logits via a decoder, $\mathbf{z} =$

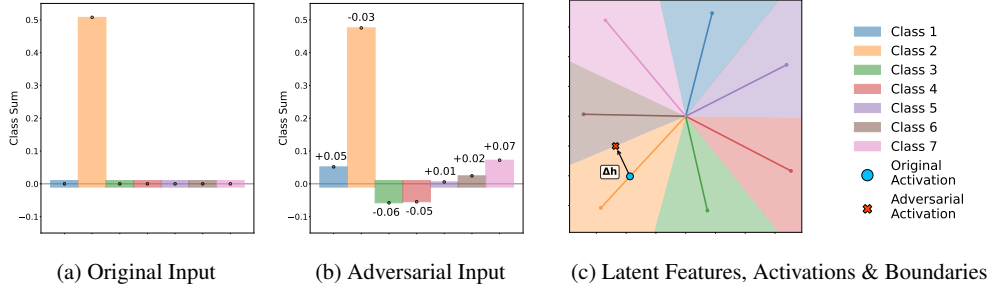


Figure 1: **An adversarial attack exploiting superposition geometry** ($k = 7, m = 2$). (a) The original sample. (b) The adversarially perturbed sample, whose ground truth remains unchanged while the model prediction changes. The sign and magnitude of each input perturbation component are determined by the configuration of latent representations. (c) The original and adversarial sample in activation space. The arrows are the column vectors of $\mathbf{W}_e$, the latent representations of the input features.

$\mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary setup uses cross entropy (CE) loss without ReLUs or biases, because this simple configuration makes the induced superposition geometry easier to analyse while the softmax limits interference by exponentially suppressing contributions from interfering latent features. To check that the conclusions are not tied to this exact loss or nonlinearity choice, we also validate them in a second setup with mean squared error (MSE) loss, ReLUs, and biases. Across both variants, the representational challenge is to compress information from $d = k \times p$ dimensions to $m < k$ dimensions, and then recover enough information to identify the class with the largest sum.

Although non-synthetic tasks involve nuanced features that must be discovered and interpreted, this setup supplies ground-truth class labels for the latent features. We can therefore compute the true class after any perturbation by recalculating group sums, and define an AEx as an input satisfying both of the following conditions:

1. The model’s prediction under perturbation changes (*e.g.* from class A to B)
2. The true class remains unchanged (*e.g.* still class A)

This separates genuine AExs from perturbations that actually change the ground truth (*i.e.* change which group has the highest sum). Attacks are created using ℓ_∞ - and ℓ_2 -norm PGD.

Evaluation metrics. We interpret the columns of the encoder weight matrix $\mathbf{W}_e$ as the latent representations $\mathbf{v}_i$ for each input feature x_i . Since input feature $x_i^{(j)}$ contributes exclusively to class j , these $\mathbf{v}_i$ (for $i \in \text{class } j$) align. This direct correspondence between input features and class-aligned latent directions lets us trace, without additional interpretability assumptions, how perturbations to the inputs of a specific class, $\mathbf{x}^{(j)}$, map into latent-space changes. For successful PGD-generated AExs $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we measure:

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? An attack must move a sample across a decision boundary to change its class, but the general question is what determines the required input perturbations. Two contrasting intuitions are possible: the *feature intuition* suggests increasing input features of the target class, while the *bug intuition* treats adversarial perturbations as effectively unpredictable from feature representations.

Fig. 1 shows a representative AEx in our setting: an input from Class 2 (orange) is perturbed, with the shown IPP, so that the model classifies it as Class 6 (brown). The perturbations do not simply increase target-class features, so the feature intuition is insufficient. They are also not arbitrary: δ

follows a consistent pattern determined by the configuration of latent features. This supports a third, more broadly applicable intuition: *attacks are mediated by latent geometry*.

Quantitatively, the IPP is strongly correlated with latent attack alignment. The sign and magnitude of perturbations $\delta^{(j)}$ track how each class’s latent representation $\mathbf{v}^{(j)}$ aligns with the latent attack vector $\Delta\mathbf{h}$. Positively aligned representations ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) have their input features proportionally amplified, while negatively aligned representations are attenuated. Thus, Fig. 1(b) illustrates input perturbations determined by the dot products in Fig. 1(c), which move the sample across the decision boundary.

To evaluate whether PGD attacks specifically exploit superposition geometry, we compare PGD-generated attacks with theoretically optimal perturbations that we show leverage superposition (derived in Sec. 3.2). For each configuration (k, m) , we train 5 models with different random seeds and generate 1000 PGD attacks per model, then compute the cosine similarity between each successful perturbation and the corresponding optimum. We define a random baseline by generating perturbations with matching ℓ_2 norm and measuring their similarity to the theoretical optimum. Tab. 1 shows near-perfect alignment between PGD and optimal attacks across multiple dimensionalities. We filter out samples with ℓ_2 norm less than ϵ to ensure meaningful perturbations. One-sample t-tests comparing observed similarities against the random baseline yield $p < 10^{-10}$ for all configurations, showing that PGD attacks systematically match theoretical predictions rather than arising by chance. Because the optimal attacks leverage interference and PGD attacks align closely with those optima, we conclude that PGD exploits superposition interference in this controlled regime.

Finding: Adversarial attacks systematically exploit interference between superposed features. Successful PGD attacks are predictable from the specific superposition geometry, rather than random.

Do input correlations determine latent feature geometry? We next ask whether correlations in the input data determine the geometric arrangements of latent features. By systematically manipulating correlation structure in the training data, we test how distributional structure constrains the degrees of freedom in learned representations. We introduce three correlation patterns to vary this constraint strength. Under the *uncorrelated* condition, features are sampled independently as in the previous subsection. For *paired* correlations each feature in class i is coupled with a feature in class j , such that when one activates, so does its pair. For *global* correlations, each sample has a random phase that determines a cyclic activation pattern: classes adjacent in index have similar activation probabilities and thus frequently co-activate, while classes further separated rarely activate together.

Fig. 2 displays how these correlations shape the arrangement of $\mathbf{v}^{(j)}$. We quantify geometric similarity by comparing the pairwise cosine similarity matrices between all feature pairs in each model, then measuring the correlation between these matrices across different random seeds. Higher correlation indicates more similar geometric arrangements. For each correlation condition, we tested 25 seed pairs per condition. For the $m = 2, k = 7$ setup, geometric similarity was 0.23 ± 0.14 (uncorrelated), 0.47 ± 0.32 (paired), and 0.88 ± 0.06 (global) (mean $\pm$ std). The results show a monotonic relationship: uncorrelated data permits highly variable geometries across seeds, paired correlations partially constrain the arrangements, and global

Table 1: Alignment between PGD-discovered attacks and theoretically optimal perturbations across superposition configurations. Results show mean $\pm$ std over 1000 attacks per condition. All p -values are below 10^{-10} .

k	m	Cosine Sim. (PGD vs Theory)	Cosine Sim. (Random Baseline)
6	2	0.97 ± 0.02	0.00 ± 0.02
30	10	0.96 ± 0.00	0.00 ± 0.01
90	30	0.92 ± 0.00	0.00 ± 0.00

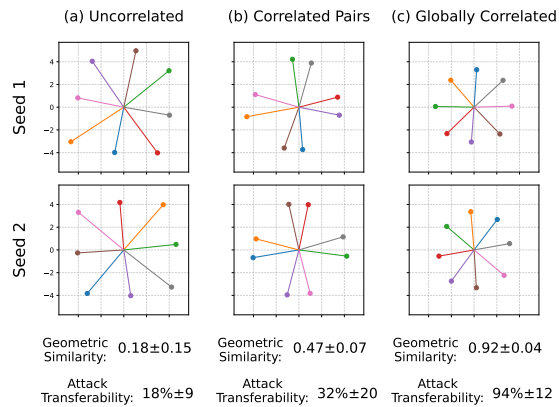


Figure 2: Stronger input correlations create more consistent latent geometries between initialisations, increasing attack transferability from 18% (uncorrelated) to 94% (global). Error bars show standard deviations.

correlations produce near-identical geometries. Statistical significance was assessed using pairwise two-sample t-tests between all three modes with Bonferroni correction ($p < 10^{-3}$ for all comparisons, $\alpha = 0.05$ corrected for 3 comparisons). Results across various (k, m) configurations are provided in Sec. C.3.

The mechanism is *interference avoidance*: frequently co-activating features are arranged to minimise mutual interference. Stronger correlations impose more constraints on the representational solution: uncorrelated data allows any arrangement that separates classes, paired correlations fix relative positions of coupled features, and global correlations determine a unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. Stronger correlations reduce geometric degrees of freedom, causing different initialisations to converge to similar arrangements.

Does shared geometry explain attack transferability? Having established that input correlations induce consistent latent geometries across model initialisations, we now test whether similar interference patterns explain attack transferability. If attacks are tailored to specific interference patterns, they should transfer successfully primarily between models with similar geometric arrangements.

Using the same seed pairs as above, we generate adversarial attacks on source models and evaluate their success when applied to target models within each correlation condition (25 transfer measurements per condition). All models achieve $> 95\%$ clean accuracy. Transfer rates correlate strongly with geometric similarity. For the $k = 7, m = 2$ configuration (Fig. 2): globally correlated data yields $94\% \pm 12\%$ attack transfer (mean $\pm$ std), uncorrelated data shows only $18\% \pm 9\%$ transfer, and paired correlations produce intermediate results of $32\% \pm 20\%$ transfer.

Each perturbation component amplifies or suppresses features, creating constructive interference that pushes representations across decision boundaries. When the same perturbation is applied to a model with different feature arrangements, features that previously constructively interfered can instead cancel out or interfere destructively, causing the attack to fail. Transferability therefore reflects shared representational geometry, not merely shared labels or task structure.

Finding: Attack transferability is governed by shared interference patterns.

Does reducing superposition suppress these attacks? If superposition creates vulnerability through interference, then reducing superposition should limit this particular adversarial mechanism. We test this with three experiments that reduce or isolate interference. First we set $m = k$, and the network learns to represent each class direction $\mathbf{v}^{(j)}$ orthogonally. In this configuration, any perturbation that changes the model’s prediction also changes the ground truth class. Moving a sample from class A to class B requires making the sum of class B features exceed class A’s, genuinely transforming it into a class B sample. We find zero successful adversarial examples across 1000 attempts at all ϵ values tested. Second, we isolate superposition’s effect from capacity by fixing m and varying k . Robust accuracy decreases monotonically with superposition pressure k/m (App. Tab. 5), demonstrating that vulnerability scales with interference degree. Third, we test whether attacks focus on superposed features by constraining one class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others during training. When generating attacks between classes that remain in superposition (e.g. from class a to b where $\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs corresponding to the orthogonal class remain unperturbed (Fig. 3).

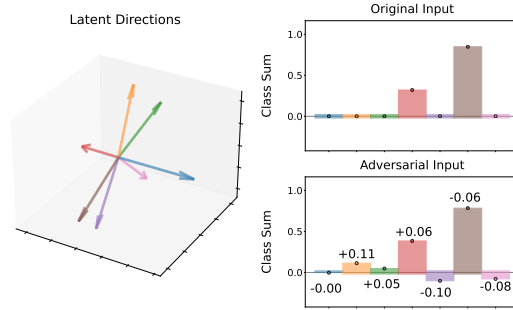


Figure 3: An adversarial attack (from class 5 to class 3) does not perturb the input features for a class represented orthogonally (class 1), isolating the role of superposed features.

Finding: Adversarial attacks allocate their budget toward features participating in superposition.

3.2 FORMAL ANALYSIS

We now formalise the mechanism underlying these empirical observations. In this linear setting, we can derive exact relationships between superposition interference and adversarial vulnerability, moving from observed correlations to a sufficient mechanistic account. We adopt the notation from our experimental setup, where $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition with columns $\mathbf{v}_i$, and the decoder corresponds to the encoder transposed (as empirically observed). The logit for class j is $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in Sec. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This derivation shows how superposition can create adversarial vulnerability in any setting matching these assumptions. Each input feature i is perturbed in proportion to its interference with the class representations. Under superposition, non-orthogonality allows semantically unrelated features to influence a class decision, and adversarial perturbations exploit these dependencies to manipulate outputs.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q} \mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in Sec. B. $\square$

Together with our empirical findings, these propositions establish a **mechanistic pathway** for the class of models satisfying the stated representation assumptions. Specifically: (i) input correlations constrain feature arrangements in superposition, (ii) these geometric arrangements determine interference patterns, (iii) interference patterns dictate optimal perturbations via $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1), and (iv) shared geometric constraints yield similar interference patterns across models, enabling transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

4 ATTACKS IN REALISTIC MODELS

We next test whether the signatures observed in the controlled setting persist in a more realistic vision-model setting, using a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009) (and ResNet-18 (He et al., 2016) Sec. D.2.1). By introducing a bottleneck layer directly before the final classification output, we induce superposition among latent class features while keeping the dataset, architecture family, and attack procedures standard. We then examine: (i) how the degree of superposition impacts attack transferability and robustness, (ii) whether a consistent arrangement of the superposed representations emerges across initialisations, and (iii) whether the attacks exploit feature interference.

Setup. We train our ViT architecture (6 layers, patch size 4, residual stream $d = 512$) on CIFAR-10 across five random seeds. After training, we replace the original classification head with a bottleneck consisting of a linear encoder that compresses the pre-classification activations to an m -dimensional space, followed by a linear decoder that expands back to the $k = 10$ CIFAR-10 classes. This bottleneck structure, trained with CE loss and no ReLU (mirroring the setup in Sec. 3), forces the k class representations into m -dimensional superposition. During training of the bottleneck layer the weights of the base ViT are frozen. We vary the bottleneck dimension $m \in \{2, 3, 5, 10\}$ to control the degree of superpositional compression. AExs are generated for each bottlenecked model using both ℓ_∞ - and ℓ_2 -norm PGD. We employ a small step size and a large number of iterations, terminating

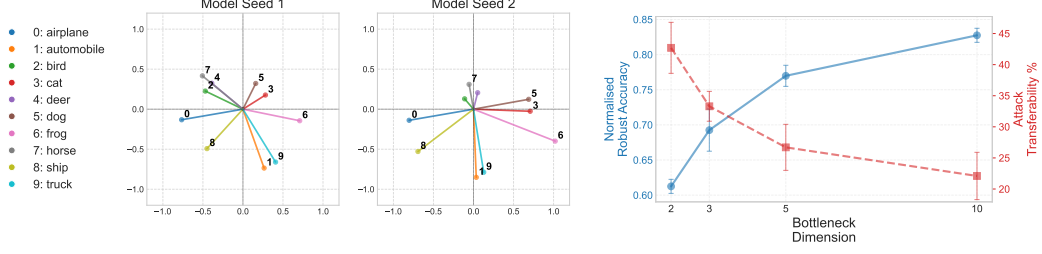


Figure 4: Left: CIFAR-10 class representation structure remains similar between models across different seeds. Right: Attack transferability and robust accuracy as bottleneck dimension is increased, varying the degree of induced superposition.

when an attack successfully changes the model’s prediction (full details in Sec. D.1). Robust accuracy is normalised by the model’s clean accuracy to account for performance variations caused by different bottleneck capacities. Attack transferability is measured as defined in Sec. 3.

Results. The mean clean accuracy of the base models is 81%. As expected for compressed classifiers, the clean accuracy of the bottlenecked models decreases with compression: 54%, 67%, 76%, and 81% for $m = \{2, 3, 5, 10\}$.

We make three observations that mirror the toy-model results, indicating that the mechanism is not confined to the synthetic task. First, Fig. 4 (right) shows that as the bottleneck m decreases (*i.e.* superposition pressure increases), (i) normalised robust accuracy decreases, and (ii) attack transferability across different model initialisations increases. The decrease in robust accuracy with increasing superposition is consistent with denser packing of class representations, which makes it easier for small input perturbations to cause widespread interference across compressed features. The increased transferability follows from our Sec. 3 finding that stronger superposition reduces the degrees of freedom in potential feature geometry. With a more constrained representational space, the network has fewer viable geometric arrangements for its class features. Different model initialisations therefore converge to more similar representational geometries and share more interference patterns, producing greater attack transferability. See Sec. D.1.1 for results across perturbation magnitudes.

Supporting this interpretation, Fig. 4 (left) shows that the geometric arrangement of the m -dimensional class representations within the bottleneck exhibits consistent ordering and clustering across models. As in Sec. 3, this consistency suggests that data correlations guide the formation of a superposed geometry. Notably, these groupings often reflect semantic similarities between classes (*e.g.* ‘cat’ and ‘dog’ representations clustering together). Finally, when AExs generated for a bottlenecked model are run through the corresponding base ViT, they produce similar relative logit changes (*i.e.* $\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$). This indicates that both models respond consistently to the same perturbations: the perturbations are not solely an artefact of the bottleneck, although without the increased compression they do not lead to misclassifications.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLINESS

The preceding sections show that feature superposition can *suffice* for adversarial vulnerability. We now ask whether it is *necessary* by studying a setting where the relevant features are represented orthogonally, not in superposition, yet the model remains vulnerable to attacks. This broadens the paper’s scope from a single vulnerability source to a representation-centred approach to robustness analysis. We ask three questions: *how do such attacks work mechanistically, can this understanding be used to construct attacks directly, without gradients, and does naive robustness certification alter the underlying failure mode or merely increase required attack budgets?*

Setup. We train a linear encoder + MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$. As detailed in Sec. E and Nanda et al. (2023), the model discovers a trigonometric algorithm to solve this task. As part of this learned algorithm¹ the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. The MLP then

¹We refer to the specific computation executed by the network to solve the task as the *model’s learned algorithm*. As this algorithm depends on extracting precise values from learned feature representations (frequencies), it remains *brittle* in the face of small perturbations to the bases. We term this the *algorithmic brittleness*.

composes $\cos((a+b)\omega_k)$ via sum-angle identities to obtain 100% accuracy. Crucially, these features are represented *orthogonally* (i.e. not in superposition). We generate AExs by relaxing the one-hot constraint and perturbing only input a using PGD, moving inputs off the training manifold.

Attacks exploit the learned key frequencies. Fourier analysis of successful perturbations δ (App. Fig. 6) reveals that the dominant frequencies precisely match the learned algorithm’s key frequencies, ω_k . 0% of attacks transfer to models with different ω_k , e.g. those trained from different seeds. The frequency-aligned perturbations cause misclassification by altering the sinusoidal values read from the encoder features. The perturbation δ transforms one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$. Contributions from the now non-zero inputs corrupt the sinusoidal values, disrupting the MLP’s subsequent computations and leading to misclassification. Because the learned algorithm uses the constructive interference of each ω_k at the logits, disrupting a single ω_k is insufficient. Successful attacks must find δ that substantially alters multiple frequency components. We leave a detailed intuition of the mechanism to Sec. E.1

Analytic construction of attacks. Using this mechanism, we can construct gradient-free, *informed* perturbations formed from composite sinusoidal waves of matching ω_k . On $0 + 0 \bmod P$, these attacks succeed with $\|\delta\|_2=0.22$, close to PGD (0.17, $\approx 1.3\times$ the PGD budget) and far below naive alternatives (single-index 0.97, ℓ_∞ noise 2.04, uniform 6.20). This provides empirical evidence that the identified mechanism is the one being attacked, and shows that representation-aware analysis can yield concrete attack recipes rather than only post-hoc explanations.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). For fixed attack budgets, the *robust* accuracy improves from pre-robustified $\rightarrow$ post-robustified as follows: 0% $\rightarrow$ 85% at $\epsilon_\infty = 5e-3$, 0% $\rightarrow$ 96% at $2e-3$, and 76% $\rightarrow$ 100% at $1e-3$. However, when attacked with PGD, the robust model still fails via the same frequency-based brittleness, but requires larger perturbation norms. Certification therefore raises budgets without removing the specific fragility itself, motivating defences that target identified semantics (here, frequency manipulation) rather than relying solely on broad norm bounds.

Relation to the paper’s thesis. This section makes two points that integrate with the paper’s broader narrative. First, superposition is *sufficient but not necessary*: orthogonal representations can still be vulnerable to attack. Second, viewing AExs through the lens of *architecture-semantics interaction* is actionable across mechanisms: it can guide both the design of attacks (via representation-informed perturbations) and the development of defences.

Although less central here, we note a shared dependence on sparsity assumptions: upstream violations of sparsity (e.g. noise or superposed features) can deliver effectively ‘pre-perturbed’ inputs to downstream orthogonal circuits, activating the same frequency-based failure.

6 RELATED WORK

Superposition & latent representations. Superposition research identifies feature interference mechanisms that can appear across model families and tasks: Nanda (2023) distinguish *representational* and *computational* superposition, while Gurnee et al. (2023) differentiate *alternating* and *simultaneous* interference. Data correlations shape these arrangements in several settings: pairs of correlated features become orthogonal (Elhage et al., 2022), correlations can drive superposition formation (Chan, 2024), and semantic clustering can emerge from correlated structure (Prieto et al., 2025a). Beyond superposition, representation geometry is shaped by spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023), and optimisation objectives (Casper, 2023). Sparse autoencoders (SAEs) have been applied to vision models to disentangle features in superposition (Lim et al., 2024; Joseph et al., 2025).

Adversarial vulnerability. The adversarial-robustness literature often follows an attack-defence dichotomy. Attacks (Szegedy et al., 2014) use small perturbations to maximize model error, including FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018). Adversarial robustness describes maintaining performance under such attacks. Defences include adversarial training (Madry et al., 2018) and certified training (Wong & Kolter, 2018; Palma et al., 2024), which provide guarantees against perturbations within specified input bounds.

The role of input & latent features on adversarial robustness. Prior work attributes vulnerability to predictive but brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019), or to learning shortcuts

(Li et al., 2023). Perturbation characteristics also vary by dataset (Maiya et al., 2021). Elhage et al. (2022) suggest that superposition may link to AExs, though Casper (2023) debates this connection. A tension between representational efficiency and adversarial robustness has been observed (Barsbey et al., 2025). Zhang et al. (2021) show that adversarial perturbations push latent representations across decision boundaries, and Ganeshan et al. (2019) find that PGD targets final layers and propose feature-based attacks. Transferability is limited by representation discrepancies between models (Li et al., 2023; Wang et al., 2024); decorrelating features reduces transfer (Wiedeman & Wang, 2022), aligning with our finding that shared interference geometry can govern transfer.

7 DISCUSSION & CONCLUDING REMARKS

We demonstrate that adversarial attacks can exploit interference patterns arising from the geometry of superposed features in NNs. Across controlled toy models and bottlenecked ViTs, our experiments show that data properties – namely correlations and sparsity – can induce distinct superposition geometries, which in turn create predictable adversarial vulnerabilities. These geometric arrangements and their interference patterns explain phenomena such as attack transferability and class-specific susceptibility within the studied conditions. This perspective frames adversarial vulnerability as a possible consequence of efficient representational compression, rather than only as an optimisation flaw or an intrinsic property of non-robust input features.

Limitations & future work. Our insights stem primarily from simplified or engineered settings in which relevant representation structure is observable or deliberately induced. While our analyses offer an initial mechanistic lens for attacks in larger models, the broader adversarial robustness literature suggests that precise vulnerability mechanisms likely vary by attack type, architecture, task, and domain. For example, constrained attack types (*e.g.* universal perturbations, jailbreaks, patch attacks) may be unable to activate the specific feature combinations required to exploit superposition interference.

Future work includes quantifying the ‘adversarial cost’ of superposition, clarifying its connection to accuracy-robustness trade-offs, and studying how robust training reshapes superposition geometry. Our present evidence comes from settings with class features in engineered or analytically tractable superposition, whereas real networks can involve interference among unknown, unlabelled features across multiple layers. We plan to use SAEs (Bricken et al., 2023) to tackle these challenges in large scale networks, with preliminary results in Sec. F indicating this is a promising direction.

ETHICS STATEMENT

Because this work is primarily mechanistic and explanatory, we do not anticipate specific negative impacts or immediate societal consequences beyond those already associated with established adversarial-robustness research. Our experiments use only publicly available data (CIFAR-10) and synthetic or algorithmic toy models, involving no human subjects or personally identifiable information. The gradient-based attack methods used in the paper are already well established in the literature. Our work does not introduce qualitatively new attack capabilities; instead, it provides a broader theoretical account of existing phenomena within the studied settings. We have no conflicts of interest to declare. By clarifying how superposition and other representation-level mechanisms can relate to adversarial vulnerability, we hope to inform future research that balances model capability with robustness and contributes to more trustworthy AI systems in safety-critical applications.

REPRODUCIBILITY STATEMENT

We include source code for all experiments in the supplementary materials. The toy model experiments (Sec. 3) include architectural details, training procedures, evaluation metrics, and data generation processes. For the ViT experiments (Sec. 4), architecture and training procedures are provided, with further details in Sec. D.1. The modular arithmetic experiments (Sec. 5) specify the network architecture, data encoding, and learned trigonometric algorithm, with additional details in Sec. E.2. Experiments were conducted on an AMD Ryzen Threadripper 3970X 32-Core with 256GB RAM and RTX 3090. Additional ResNet-18 and SAE experiments are documented in Appendices Sec. D.2.1 and Sec. F, respectively.

LLM USAGE STATEMENT

This work utilised LLMs to assist with prior literature search, automate LaTeX table generation from CSV files (all numerical values were verified), and suggest alternative phrasings to improve manuscript clarity and framing. All substantive research contributions and conclusions remain the original work of the authors.

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APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We use PGD to generate AExs, following a standard iterative attack procedure (Madry et al., 2018). For untargeted attacks, it maximises the classifier’s loss function $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$; for targeted attacks, it minimises $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$ by the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathcal{S}} \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

(+ for maximisation, − for minimisation). Here $\mathbf{x}'^{(0)}$ is an initial perturbed input, α is step size, $\mathbf{g}_k$ is the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ (y_{class} being y_{true} or y_{target}), and $\Pi_{\mathcal{S}}$ projects onto the allowed ϵ -ball $\mathcal{S} = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$, with input domain clipping. We use the following ℓ_p norms:

1. ℓ_{∞} : Constraint $\|\delta\|_{\ell_{\infty}} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. $\Pi_{\mathcal{S}}$ clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. $\Pi_{\mathcal{S}}$ rescales $\delta = \mathbf{x}' - \mathbf{x}$ if $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

To clarify when representation geometry determines adversarial perturbations, we prove that optimal perturbations weight each input dimension by how much its corresponding feature aligns with the path to the decision boundary. The derivation applies to our linear models without ReLU activation functions.

Consider input $\mathbf{x} \in \mathbb{R}^d$ encoded via $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ to latent representation $\mathbf{h} \in \mathbb{R}^m$, where the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ form an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ ($m < d$). For our argmax task, we empirically observe that the trained encoder-decoder pair has the decoder as the transpose of the encoder: $\mathbf{W}_d = \mathbf{W}_e^{\top}$. The logit for class j is therefore computed as $z_j = \mathbf{v}_j^{\top} \mathbf{h}$, where $\mathbf{v}_j$ is the j -th column of $\mathbf{W}_e$. The predicted class is $\hat{y} = \arg \max_j z_j$. Input perturbations map to latent perturbations as $\Delta \mathbf{h} = \mathbf{W}_e \delta$, where $\delta = (\delta_1, \dots, \delta_d)^{\top}$ are the perturbation coefficients.

The decision boundary between classes j and k is the set of points where their logits are equal: $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$, or equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} = 0\}$. The vector $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to this pairwise decision boundary and points in the direction that maximally increases the margin $z_k - z_j$. We briefly recall the propositions before providing the corresponding proofs.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise movement from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^{\top} \mathbf{n}$$

Proof. To move a sample from being classified as j to being classified as k , we need to maximise the logit margin $z_k - z_j$. Under perturbation δ , the new margin becomes:

$$z'_k - z'_j = \mathbf{v}_k^{\top} (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^{\top} (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta = \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^{\top} \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

□

Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

This reveals the mechanism by which superposition can create adversarial vulnerability under the linear assumptions above. Each input feature i is perturbed proportionally to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$ —the differential interference between feature i and the class representations. Under superposition, non-orthogonality means that even semantically unrelated features can have non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$, creating exploitable interference patterns. The multi-class setting broadens this vulnerability, since with k classes there are multiple possible pairwise boundaries, each creating a distinct interference pattern. Adversarial perturbations leverage these cross-feature dependencies through their interference with the class representations. This explains the vulnerability observed empirically: attacks succeed not by directly increasing target-class features, but by exploiting the web of interference created by superposition.

Prop. 1 characterises the optimal perturbation direction for moving from class j to k , providing the gradient for maximising the margin $z_k - z_j$. When intervening classes exist (where $z_j < z_i < z_k$ for some class i), following this gradient might cause the model to predict i before reaching target k . Iterative methods like PGD handle this by recomputing gradients at each step—the global attack path emerges from repeated local decisions rather than a single optimisation. Furthermore, while our analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ based on our empirical observations, the framework extends to arbitrary decoders. In the general case, with decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ having rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, capturing interference between encoder directions and decoder weights.

Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (i.e., $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. □

Prop. 2 is used to explain why attacks transfer between models with similar training regimes. When models learn feature representations that differ only by orthogonal transformation—essentially the

same geometric structure in different orientations—they share identical vulnerability patterns in input space. The orthogonal transformation preserves all inner products between features, maintaining the interference patterns that attacks exploit. This provides an explanation for our empirical observation that models trained on data with the same correlations exhibit high attack transferability: they discover similar feature geometries up to rotation, leading to shared adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section provides supplementary details, extended results, and further discussion for the toy model experiments in the main paper. We present model accuracies across a wider range of parameters than shown in the main text, clarifying how model capacity and data characteristics such as sparsity influence the learning process and the conditions under which feature superposition appears. We then provide additional visual examples corresponding to Fig. 1, illustrating the same attack mechanism under varied conditions.

C.1 HYPOTHESES TESTING FRAMEWORK

We explicitly state the hypotheses corresponding to the three research questions in Sec. 3.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We test these hypotheses through controlled experiments that vary representation geometry while preserving the task definition:

- $H_1(1)$: We measure the input perturbation profile alignment with a class’s latent representation and the latent attack vector.
- $H_1(2)$: We systematically vary input correlations and measure resulting geometries.
- $H_1(3)$: We quantify transferability rates across models with varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

The toy model experiments in the main paper primarily use low-dimensional settings for conceptual clarity. To demonstrate the model’s behaviour across a broader parameter range, this subsection details the classification accuracies achieved by the CE toy model. These results are presented across varying hidden layer size (h), number of classes (k), number of features, and levels of sparsity (S), providing insight into when the models learn to represent features in superposition. The sparsity level is the probability that any individual input feature $x_i^{(j)}$ is set to zero, with higher values of S indicating greater input sparsity. We report results as a function of feature density $1 - S$, the probability that a feature is non-zero. Tab. 2 and Tab. 3 provide context on the model’s performance limits and its ability to learn latent representations in superposition.

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 displays results for the experiments addressing *Do Input Correlations Determine Latent Feature Geometry?* across a wider range of classes (k) and hidden dimension size (m).

Table 2: Classification accuracy of the CE toy model with a fixed bottleneck dimension ($m = 2$) across various numbers of classes (k), total input features (features = $k \times 3$), and input feature densities ($1 - S$). These results illustrate how input sparsity controls performance degradation as the number of classes to be superposed within a constrained latent space increases.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model for varying numbers of classes (k), total input features, bottleneck dimensions (m), and input feature sparsity levels (S). This table reports different numbers of features per class (p).

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

Table 4: Geometric similarity results across correlation types and superposition configurations, showing how correlation structure constrains learned feature geometry.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

How might scaling networks affect adversarial vulnerability? Empirical evidence suggests that larger models tend to be somewhat more robust to adversarial attacks, but the effect is weak. When adversarial training is employed, clearer scaling trends emerge, although improvements remain largely specific to the attack type used during training rather than conferring general robustness (Howe et al., 2025).

Two phenomena are relevant when scaling models. On one hand, larger models have more capacity to represent concepts; on the other, broad tasks such as next-token prediction over internet text contain a long tail of useful concepts to capture. Thus, despite increased model capacity, superposition appears to remain prevalent even in frontier models (Lindsey et al., 2025). Supporting evidence comes from dictionary learning methods: SAEs require increasingly large dictionaries for larger models (Gao et al., 2025), suggesting that the number of represented features scales with model size. The fundamental tension driving superposition – compressing many features into limited dimensions – therefore may not disappear with scale.

Since both superposition and adversarial vulnerability persist in large-scale models, our results may remain relevant across model scales, subject to the representational assumptions made in the paper. Understanding how superposition geometry changes with scale is therefore an important future direction for mitigating vulnerability.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

Keeping the bottleneck dimension fixed helps separate superposition pressure from the confounding effect of reduced capacity. We fix bottleneck dimension $m = 2$, and vary number of classes k to isolate superposition pressure. We report the model accuracy, robust accuracy, and transferability in Tab. 5. These controls show that increased superposition pressure (k/m), independent of model capacity, drives the adversarial vulnerabilities observed in this setup.

C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

Fig. 1 demonstrates how adversarial attacks exploit interference between latent features in superposition. Here we provide further visual examples (Fig. 5) to show the same mechanism across nearby attack and bottleneck conditions. Specifically, we supplement the main text by showcasing:

- Fig. 5a shows an additional instance of the setup in Sec. 3.1 ($m = 2, k = 7$) with an ℓ_2 -norm PGD attack, demonstrating the IPP and latent space manipulations that lead to misclassification.
- Fig. 5b shows a similar setup ($m = 2, k = 7$) using an ℓ_∞ -norm PGD attack on a less sparse input.
- Fig. 5c shows an example with increased bottleneck dimensionality ($m = 3, k = 7$) and accompanying ℓ_2 -norm PGD attack. The feature vector similarity matrix used to calculate geometric similarity is also shown.

D CIFAR-10 EXPERIMENTS

To investigate whether the principles observed in the toy models extend to a more complex vision setting, Sec. 4 introduces experiments trained on CIFAR-10 (Krizhevsky, 2009) with an engineered bottleneck. This appendix section provides further details on the setup and presents extended results.

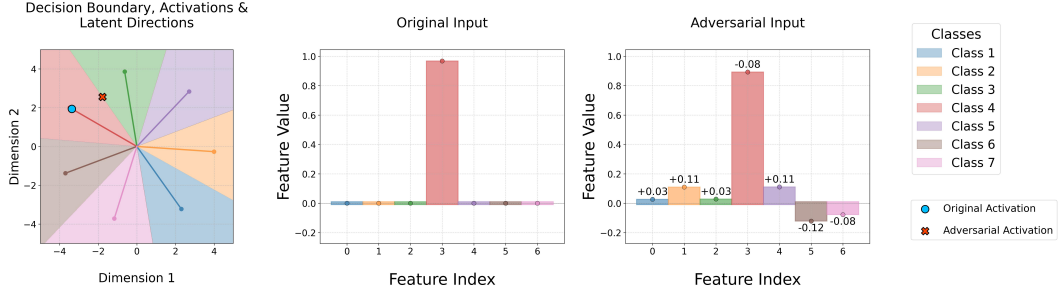
D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) architecture comprised 6 transformer layers, an embedding dimension (d) of 512, and 8 attention heads in each transformer layer. Input images from the CIFAR-10 dataset, sized at 32×32 pixels, were processed into patches of 4×4 pixels. The Multilayer Perceptron (MLP) within each transformer block had a hidden dimension of 512, and learned positional embeddings were used.

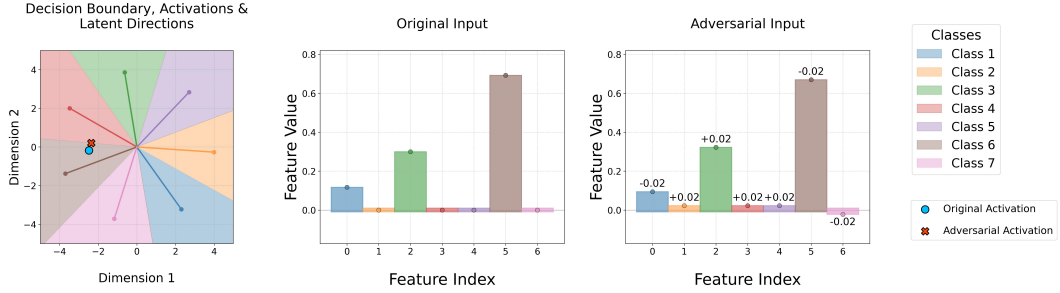
The bottleneck architecture consisted of a linear encoder followed by a linear decoder. The linear encoder projected the pre-classification activations obtained from the ViT backbone into an m -

Table 5: Performance metrics across correlation types, perturbation budgets (ε), and number of classes (k), isolating superposition pressure at fixed bottleneck dimension.

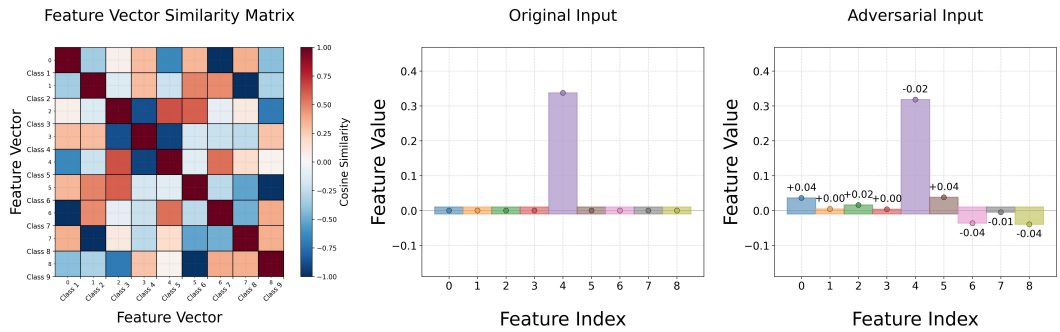
Correlation Type	Parameters		Performance (%)		
	ε	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
Fully Correlated	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0



(a) An ℓ_2 -norm attack changing the classification of an input of class 4 to class 3. The left plot shows original and adversarial activations in latent space, along with representation directions. The right plots show original and perturbed input feature values.



(b) An ℓ_∞ -norm attack changing the classification of an input of class 6 to class 4. The left plot shows original and adversarial activations in latent space relative to class latent directions. The right plots show original and perturbed input feature values.



(c) An ℓ_2 -norm adversarial attack in a 7-class setup with increased bottleneck dimension $m = 3$. The leftmost plot shows the cosine similarity matrix between pairs of latent representations for each class.

Figure 5: Visualisations of AExs in the toy model, supplementing Figure 1 from the main paper by illustrating the representation-mediated attack mechanism in activation space and input space under varied conditions.

dimensional latent space ($\{2, 3, 5, 10\}$). The subsequent linear decoder mapped these m -dimensional representations back to the $k = 10$ dimensions corresponding to the CIFAR-10 classes.

The base ViT model was trained on the CIFAR-10 dataset for 250 epochs. A learning rate of 0.001 was used with the Adam optimiser using default PyTorch parameters. The batch size was set to 512, a cosine annealing learning rate scheduler was used, the loss function was CE, and dropout was applied. Training was performed across five different random seeds to account for variability. After the base ViT model was trained, its weights were frozen. The bottleneck layer was then trained for 30 epochs with a learning rate of 0.001. For preprocessing, images undergo RandomCrop with 4-pixel padding, resize to the target size (32x32 by default), and RandomHorizontalFlip for data augmentation.

D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) shows how normalised robust accuracy varies with bottleneck dimension for one value of ϵ . This subsection expands on those findings by detailing normalised robust accuracy under PGD attacks of different strengths. Results are presented for both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) PGD attacks, providing a broader view of how the degree of induced superposition interacts with attack strength to affect model robustness.

AExs for these evaluations were generated using PGD with 100 iterations and step size (α) of 0.01. Robust accuracy was evaluated on 500 samples for each configuration. We report the mean normalised robust accuracy and standard deviation across five random seeds.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10, subjected to ℓ_∞ -norm PGD attacks of varying perturbation magnitudes (ϵ). Robust accuracy is normalised by the clean accuracy of each bottlenecked model. These results support the findings of Sec. 4, showing the same qualitative trend across ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10, subjected to ℓ_2 -norm PGD attacks of varying perturbation magnitudes (ϵ). Robust accuracy is normalised by the clean accuracy of each bottlenecked model. These results support the findings in Section 4 of the main paper, showing decreasing robustness with smaller m (increased superposition) and larger ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

We include results on attack transferability across perturbation magnitudes (ϵ) and bottleneck dimensions (m). Tab. 9 presents ℓ_2 -norm attack transferability and Tab. 8 presents ℓ_∞ -norm attack transferability.

D.2 CORRELATIONS IN CLASS FEATURES

In Sec. 3, correlations between inputs drove superposition arrangements. In the CIFAR-10 setting, the analogous driver is not raw correlations between input classes, but correlations in the representations at this point in the network. At the classification layer this is likely related to the confusion matrix – *i.e.* how each class is misclassified relative to other classes. Classes that cluster together are those the network finds inherently similar and tends to confuse. To test this, we repeat the experiment by fine-tuning the base ViT using timm/vit_base_patch16_384, which has been trained on ImageNet-21k (14 million images, 21,843 classes) and ImageNet (1 million images, 1,000 classes). After fine-tuning on CIFAR-10 and applying the same bottleneck training procedure as Sec. 4, the resulting geometry shows random ordering between initialisations with approximately equal spacing between features (*i.e.* neuron collapse Kothapalli (2023)). In this case performance is near 100%, meaning the confusion matrix is the identity, and the superposed geometry loses its structure. In contrast, models trained solely on CIFAR-10 converge to similar geometries because they share the same learned difficulty structure: the same pairs of classes are challenging to distinguish, leading to consistent superposition patterns.

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT models. Transferability is shown from a model trained with a specific ‘Source Seed’ (e.g., Seed 10) to three different target models, each trained with one of the seeds listed in the sub-header (e.g., ‘vs. Seeds 20/30/40’). The three slash-separated values in each cell correspond to transferability to these three target seeds, respectively. All models within a row share the same bottleneck dimension, m . The ‘Mean $\pm$ Std’ column averages transferability across all 12 source-target seed pairings for each (ϵ , m) configuration. These results support the claim in Section 4 that higher induced superposition can lead to more consistent latent geometries and therefore higher transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 $\pm$ 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 $\pm$ 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 $\pm$ 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 $\pm$ 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 $\pm$ 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 $\pm$ 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 $\pm$ 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 $\pm$ 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 $\pm$ 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 $\pm$ 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 $\pm$ 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 $\pm$ 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 $\pm$ 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 $\pm$ 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 $\pm$ 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 $\pm$ 3.3

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT models. The table format, detailing source model to target model transferability, mirrors that of Tab. 8; please see its caption for a full explanation. These ℓ_2 results further support the claim in Sec. 4 that higher induced superposition leads to increased attack transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

D.2.1 RESNET-18

To address concerns on architectural generalisation, we conduct the same experiments using a ResNet-18 (He et al., 2016) architecture as the base model as opposed to a ViT. We achieve a slightly higher 92% clean accuracy (compared to 89% for ViT). We observe the robust accuracy falls slightly faster for ResNet-18 than the ViT at the same ϵ values. Nevertheless, we observe similar declining trends in normalised robust accuracy and increasing transferability as bottleneck pressure increases.

E MODULAR ADDITION

Sec. 5 introduces a modular addition task to investigate adversarial vulnerability in a scenario where the relevant features are orthogonal. This section provides the detailed experimental setup, additional results, and further discussion of how representation-level analysis can be useful beyond superposition-driven attacks.

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As detailed in Nanda et al. (2023) and Prieto et al. (2025b), the network discovers and implements a trigonometric algorithm to solve this task: the encoder projects inputs onto non basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. These features are represented orthogonally, *i.e.* not in superposition, and the sinusoidal values can be read off via linear probes. The subsequent MLP applies sum-angle identities to compute $\cos((a + b)\omega_k)$ for

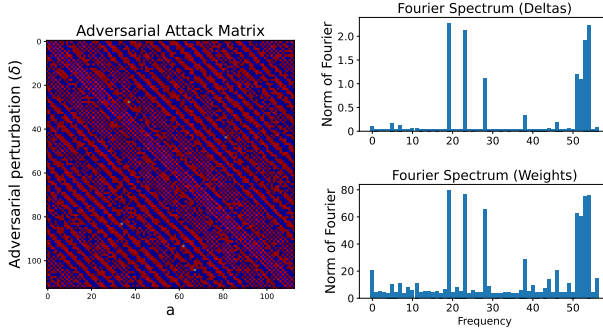


Figure 6: Left: Each column is the mean attack δ added to a to successfully attack $a + b \pmod{p}$. Right: The norms of the Fourier components in δ and $W_{MLP_{out}}$, showing alignment with the learned algorithm.

Table 10: Comparison of the minimum ℓ_2 -norm budget required to successfully attack $0 + 0 \pmod{p}$ across attack types.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

various k , which constructively interfere at the logits to select the correct class. AExs are generated by relaxing the one-hot constraint and perturbing input $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ via ℓ_2 and ℓ_∞ -norm PGD. For simplicity, only a is attacked while b is fixed.

As explained in Nanda et al. (2023), the network learns a structured trigonometric algorithm. W_e encodes integers' sine and cosine values for key frequencies, and these features are linearly decodable. Scaling the corresponding feature vector alters the encoded sinusoidal value itself, not merely its activation intensity. Integer embeddings W_e^j project onto a Fourier-like basis, creating a circular/periodic structure in the embedding space. Unlike arrangements from data correlations, this circularity arises from the sine/cosine encoding of the non-correlated input integers.

Our model learns to use approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j predominantly encodes its fourteen corresponding trigonometric features: $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$. The concatenated 200-dim. vector thus provides these ≈ 28 key trigonometric features (fourteen for a and fourteen for b) to the MLP. The MLP uses them, via sum-angle identities (e.g., $\cos(A + B) = \cos A \cos B - \sin A \sin B$), to calculate $(x_1 + x_2) \pmod{P}$. The algorithmic brittleness explored in Section 5 stems from adversarial perturbation of these key learned trigonometric features.

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113. Inputs are integer pairs (a, b) , $a, b \in \{0, \dots, 112\}$. Each integer is mapped to a 100-dimensional vector via a shared embedding $W_E \in \mathbb{R}^{113 \times 100}$. Embeddings W_E^a and W_E^b are concatenated into a 200-dim. feature vector. This vector feeds a 3-hidden-layer MLP (200 ReLU neurons/layer). The output layer provides 113 logits for the classification. The model is trained end-to-end using cross-entropy loss and AdamW (weight decay 4), achieving 100% test accuracy. AExs are generated using PGD under ℓ_2 and ℓ_∞ norms with varying ϵ . Attacks perturb the embedding W_E^a (with integer b fixed) by adding a perturbation δ to cause misclassification of $(a + b) \pmod{113}$.

E.3 ALGORITHMIC BRITTLINESS

The frequency-aligned perturbations cause misclassification by altering the sinusoidal values read from the encoder features. We use the hypothesised neuron-aligned case to give intuition: W_e would contain a row corresponding to the sine/cosine for each ω_k , whose element at column $\mathbf{w}_e^a$ contains the value $\sin(\omega_k a)$ or $\cos(\omega_k a)$. For the input a , the one-hot $\mathbf{x}_a$ selects $\mathbf{w}_e^a$, giving $\mathbf{h}_a = \mathbf{w}_e^a$. The perturbation δ transforms one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$. Now $\mathbf{h}'_a = W_e \mathbf{x}'_a = (1 + \epsilon_a) \mathbf{h}_a + \sum_{k \neq a} \epsilon_k \mathbf{w}_e^k$. Contributions from the $k \neq a$ sinusoidal terms corrupt the sinusoidal values, disrupting the MLP's subsequent computations and causing misclassification. The model's algorithm uses the constructive interference of each ω_k at the logits, so disrupting a single ω_k is insufficient because the other components strongly support the correct answer. Thus, effective attacks must find δ that substantially alters multiple frequency components.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper describes how understanding the learned trigonometric algorithm in the modular addition task allows for the analytic construction of adversarial attacks. Sec. 5 benchmarks this ‘informed attack’ against PGD and naive baselines. This section provides further details on those baseline attack methods. The baselines are:

- **PGD Attack:** Standard PGD used to find the adversary.
- **Informed Attack:** The analytically constructed perturbation, from combinations of cosines.
- **Single a :** This involved finding the minimal attack to a single input to cause misclassification.
- **ℓ_∞ Noise:** This baseline involved adding random noise δ_{noise} to the input a . Each component of δ_{noise} was sampled independently and uniformly from $[-\epsilon, \epsilon]$. The ϵ that successfully induced a misclassification is reported.
- **Uniform δ :** The perturbation vector δ is constrained to have all components equal, i.e., $\delta_j = c \cdot \mathbf{1}$, where c is a scalar. The magnitude of c is reported such that adding δ to the input causes a misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

To investigate the impact of defences on the identified vulnerability, we performed certified robust training. This aimed to make the network provably robust against ℓ_∞ -norm perturbations. The certified training method employed was an abstraction-based technique (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022), namely RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023). RSIP-IBP uses Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) and Reversed Symbolic Interval Propagation (RSIP) to compute reachable output sets for network layers, enabling formal certification of robustness within a given perturbation budget. The training loss function incorporated both a standard loss component and an RSIP loss component. Certified training was conducted for 1000 epochs, with the initial 200 epochs serving as a warm-up phase. The learning rate was 1×10^{-3} and weight decay 1×10^{-4} . The same train/test data splits were used as for the standard model training. The results of certified training across different training perturbation budgets (ϵ_{train}) and evaluated against various attack perturbation budgets (ϵ_{attack}) are summarised in Tab. 11.

Table 11: Verified robust accuracy (%) for modular addition models under ℓ_∞ -norm perturbations. The table compares a standard (pre-robustification) model with models certifiably trained using RSIP-IBP with different training perturbation budgets (ϵ_{train}). Accuracies are evaluated against various attack budgets (ϵ_{attack}). These results support the discussion in Section 5.1 of the main paper, indicating that certified training improved robustness to specified ϵ_{train} but attacks still succeeded by exploiting the same algorithmic sensitivities at higher ϵ_{attack} .

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

As discussed in Sec. 5, when these robustly trained models were subsequently attacked using PGD, successful adversarial perturbations remained mechanistically consistent: they continued to exploit the same key frequency components inherent to the model’s learned algorithm. The primary effect of certified training was an increase in the perturbation magnitude required to achieve misclassification

for epsilons against which the model was successfully certified, rather than a fundamental change in the attack vector’s alignment with the model’s algorithmic sensitivities.

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although the vulnerability explored here arises without feature superposition, it resembles superposition-driven vulnerabilities in one respect: both depend on sparsity in the input representation. This motivates our *reconciliation conjecture*: algorithmic vulnerabilities may persist in networks with orthogonal representations when sparsity assumptions are violated upstream. If earlier layers in a network produce noisy or non-sparse features, potentially due to adversarial interference, then these deviations can propagate into otherwise well-isolated downstream circuits. In such cases, circuits that operate correctly on clean inputs may nonetheless fail because the input they receive is effectively already perturbed. This provides a plausible route by which algorithm-disruption mechanisms, like the frequency attacks studied here, could be activated even in the absence of local superposition. More broadly, the interaction between upstream superposition and downstream algorithmic brittleness, as well as layer-by-layer understanding of the specific vulnerability mechanisms triggered by AExs, remains underexplored. We leave detailed investigation, grounded in empirical evidence and theoretical modelling, as a promising direction for future work.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The analyses in the main paper focus on scenarios with either explicitly defined class features (toy model) or class-level representations in a bottleneck (ViT experiments). In large practical networks, however, interference may occur among unlabelled latent features across many layers. As discussed in the future work section of the main paper, SAEs offer a promising unsupervised method for extracting more linear features from NNs (Bricken et al., 2023). This appendix section presents a preliminary investigation into using SAE-extracted features to characterise how a large scale ViT responds internally when processing AExs versus clean inputs. These initial results explore the feasibility of this approach for future work on feature-level adversarial phenomena in large models, rather than presenting a conclusive new set of findings. We first outline the experimental setup and validate the SAEs behaviour on adversarial inputs before presenting exploratory layer-wise feature difference analyses.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of the LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K model, a 12-layer ViT with $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009). We use open-source Prisma SAEs mapping the 768-dimensional activation space to a dictionary of 49,152 features ($64\times$ expansion) (Joseph et al., 2025). We examine SAEs trained on either mean patch token representations or CLS token representations, across all layers. We analyse features activating above a high threshold (0.1), to capture changes to the strongest features, and a low threshold (0.001), to include weaker but potentially informative activations. We generate 10,000 original/successfully-attacked image pairs (ℓ_∞ -norm, $\epsilon = 0.02$) for our SAE feature comparisons. This setup enables an initial exploration of how adversarial perturbations may manifest at the level of SAE features.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Because AExs are out-of-distribution inputs, and our aim is to use standardly trained SAEs to analyse them, we first conduct a basic validation. This subsection examines the impact on model accuracy and SAE reconstruction when SAEs are inserted at inference time for clean and adversarial inputs. Before interpreting differences in feature activations, we check whether the SAEs continue to process features relevant to the classification task under adversarial conditions.

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection presents exploratory visualisations comparing SAE feature activations across different ViT layers for original, ℓ_∞ adversarial, and random noisy inputs. These visualisations are intended to

Table 12: Impact on ImageNet-1k classification accuracy when inserting a pre-trained SAE (one layer at a time) during inference for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Accuracy’ refers to the ViT’s top-1 accuracy. ‘Original input + SAE’ means the clean ViT activations at a given layer are passed through the SAE and then its reconstruction is passed to the next layer. The slight ‘de-attacking’ effect (improved accuracy for ‘Adv. input + SAE’ vs ‘Adv. input’) suggests SAEs might filter some adversarial noise, but overall, the model still largely misclassifies, indicating that salient adversarial changes for misclassification are processed.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

identify potential avenues for future investigation into how adversarial attacks alter sparse feature representations. We examine:

- Activation count: Histograms comparing the number of SAE features activated above a set threshold in a layer for original, adversarial, and noisy images. This explores whether attacks systematically alter feature sparsity.
- Feature overlap: The degree of overlap (number of common activated features) between (original vs. adversarial) and (original vs. noisy) conditions. This explores whether attacks activate a distinct set of features.
- Jaccard Similarity: The Jaccard similarity between sets of activated features, offering a normalised measure of overlap.
- Mean activation: Histograms of mean activation values for features active above a threshold. This looks for changes in the intensity of activated features.
- Distinct Features: The count of unique features activated across an image set for each condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection presents a preliminary analysis of SAE features derived from CLS tokens, considering only features activated above a threshold of 0.1. Figures 7 through 10 visualise and compare activation counts, feature overlap, Jaccard similarity, and mean activations, respectively, for original, adversarial, and random noisy inputs across different ViT layers.

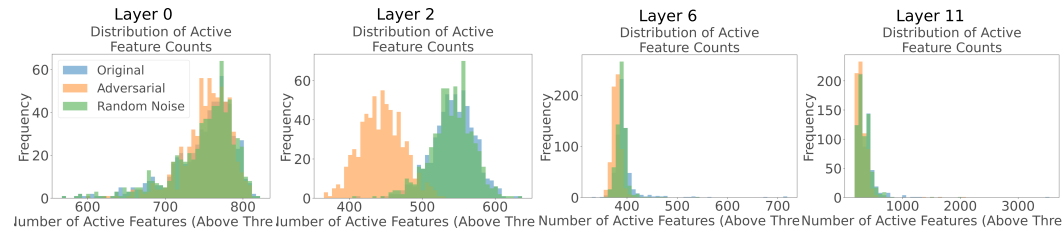


Figure 7: Histogram of activation counts for CLS token SAE features at threshold 0.1.

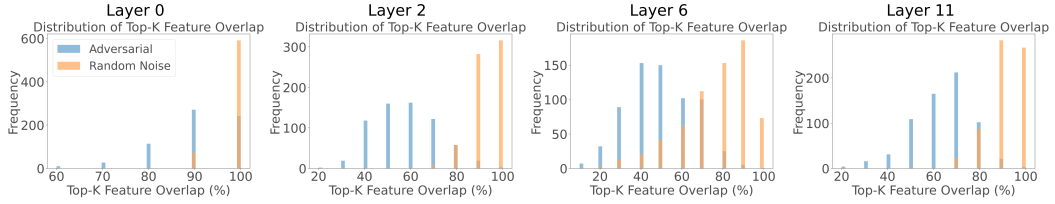


Figure 8: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features at threshold 0.1.

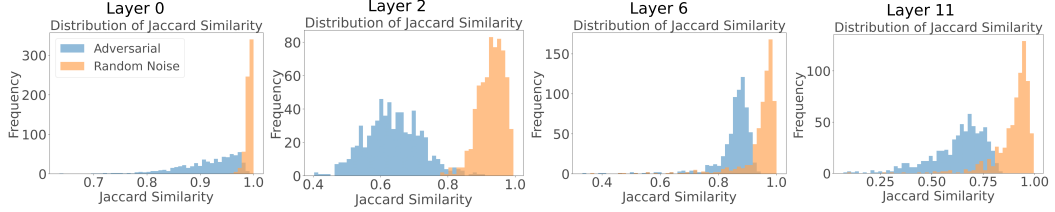


Figure 9: Comparison of Jaccard similarity for CLS token SAE features at threshold 0.1.

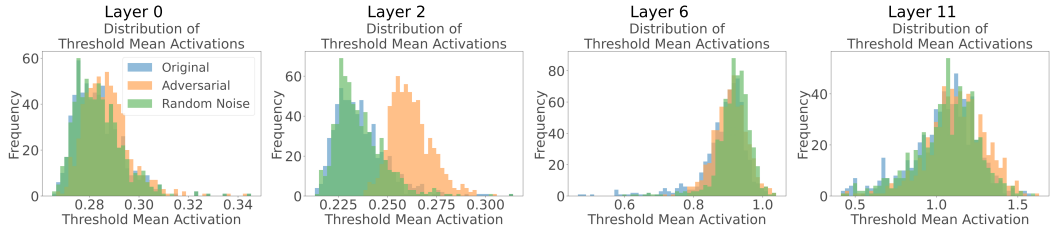


Figure 10: Histogram of mean thresholds for CLS token SAE features at threshold 0.1.

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Here, we examine SAE features from CLS tokens with a lower activation threshold of 0.001, so that the analysis is not restricted to the most intensely activated features. The subsequent figures illustrate activation counts (Fig. 11), feature overlap (Fig. 12), Jaccard similarity (Fig. 13), and mean activations (Fig. 14) across layers for original, adversarial, and noisy inputs.

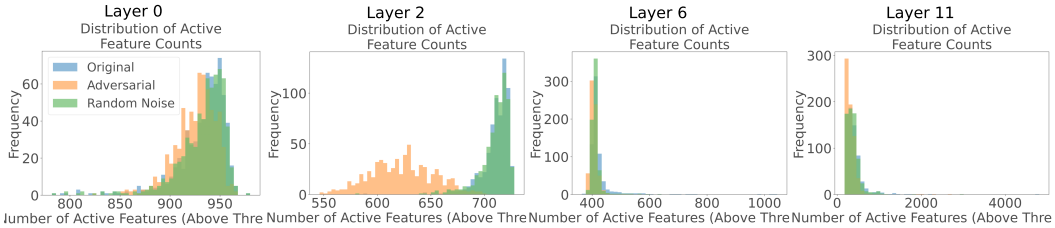


Figure 11: Histogram of activation counts for CLS token SAE features at threshold 0.001, with noise.

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection shifts to SAE features derived from the mean of patch tokens, again using a high activation threshold of 0.1. We present comparisons of activation count histograms (Fig. 15), distinct feature count histograms (Fig. 16), feature overlap (Fig. 17), Jaccard similarity (Fig. 18), and mean activation histograms (Fig. 19) for original, adversarial, and noisy inputs.

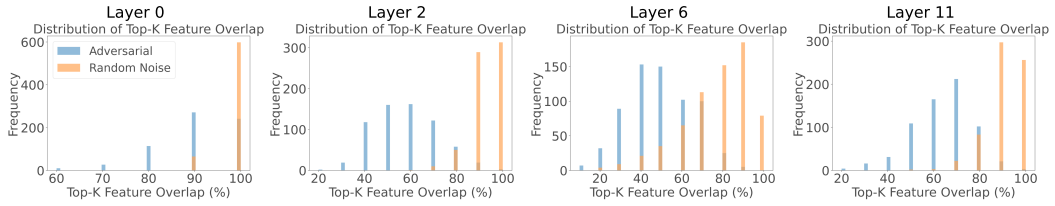


Figure 12: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features at threshold 0.001.

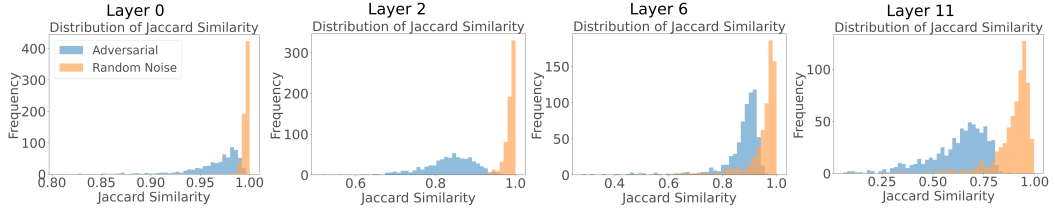


Figure 13: Comparison of Jaccard similarity for CLS token SAE features at threshold 0.001.

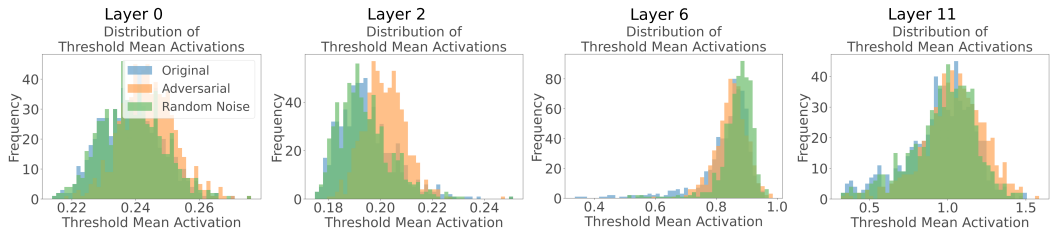


Figure 14: Histogram of mean activations for CLS token SAE features at threshold 0.001, with noise.

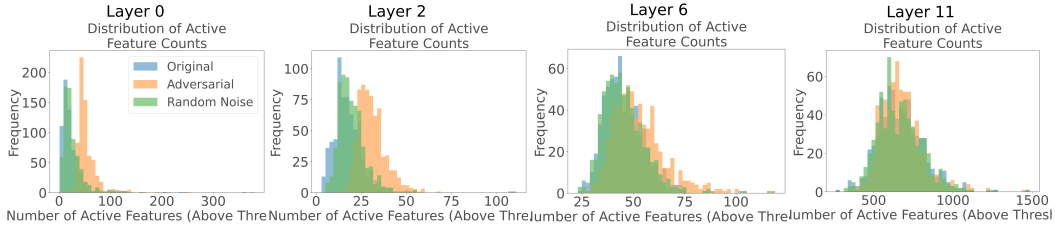


Figure 15: Histogram of activation counts for patch token SAE features at threshold 0.1, with noise.

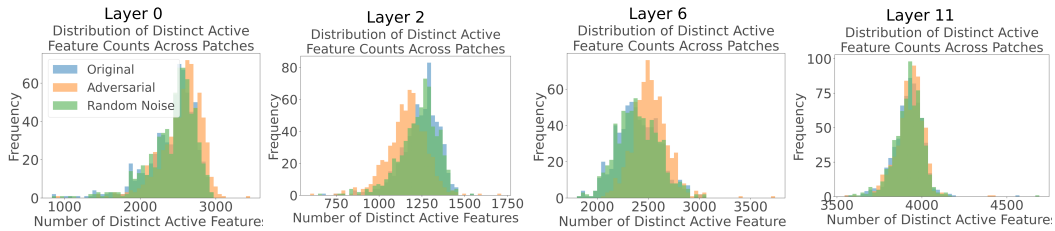


Figure 16: Histogram of distinct patch features for patch token SAE features at threshold 0.1, with noise.

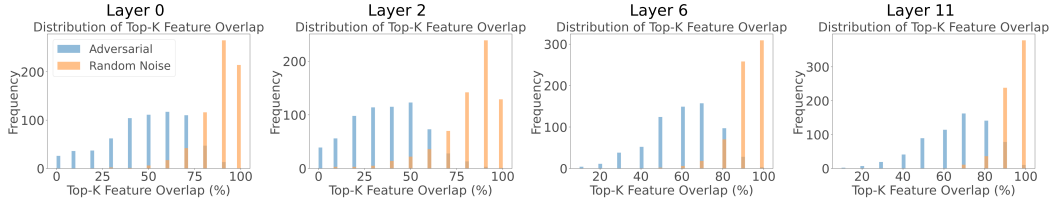


Figure 17: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features at threshold 0.1.

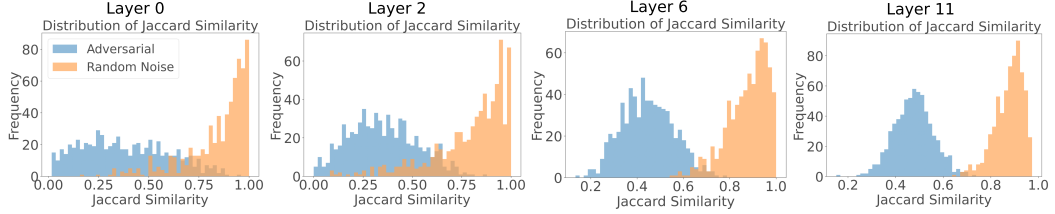


Figure 18: Comparison of Jaccard similarity for patch token SAE features at threshold 0.1.

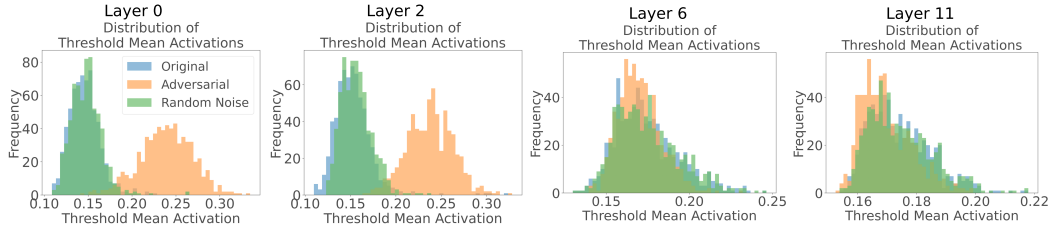


Figure 19: Histogram of mean thresholds for patch token SAE features at threshold 0.1, with noise.

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, this subsection explores SAE features from the patch token means using a low activation threshold of 0.001. The figures show activation counts (Fig. 20), distinct feature counts (Fig. 21), feature overlap (Fig. 22), Jaccard similarity (Fig. 23), and mean activation histograms (Fig. 24) for original, adversarial, and noisy inputs.

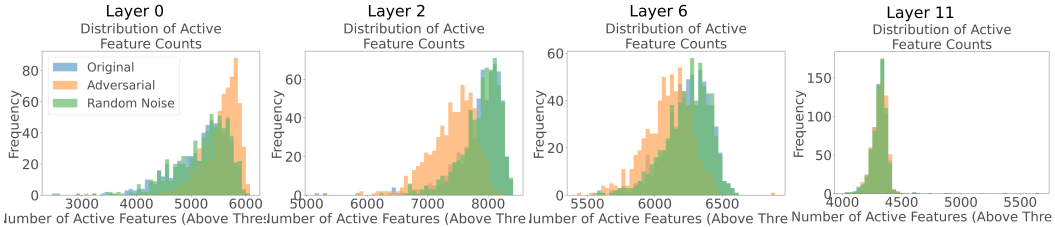


Figure 20: Histogram of activation counts for patch token SAE features at threshold 0.001, with noise.

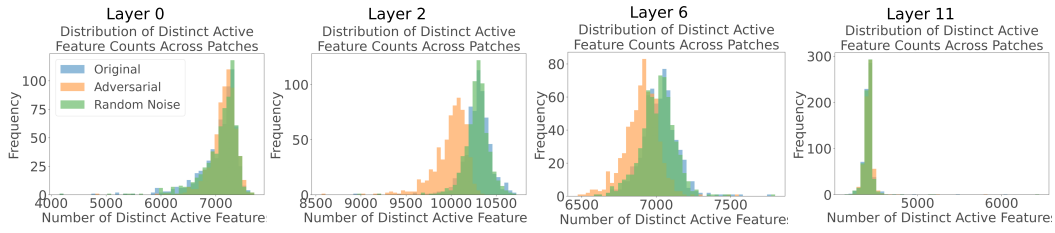


Figure 21: Histogram of distinct patch features for patch token SAE features at threshold 0.001, with noise.

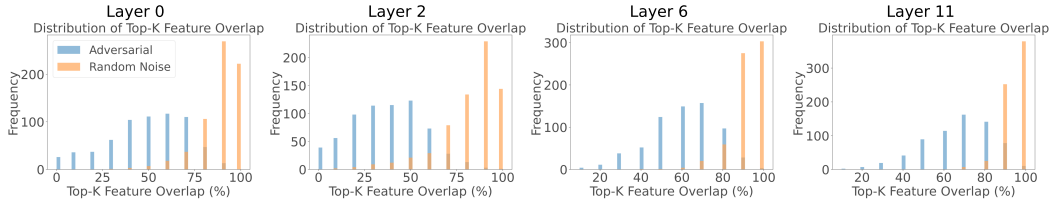


Figure 22: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features at threshold 0.001.

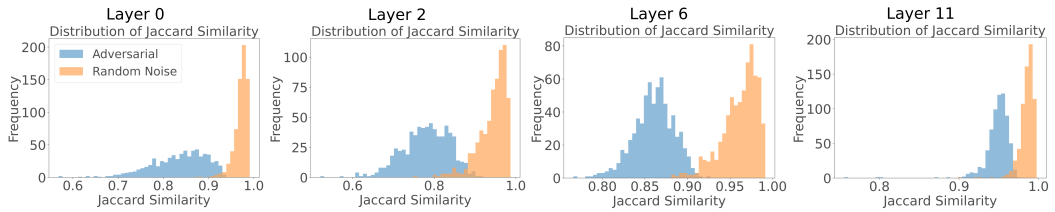


Figure 23: Comparison of Jaccard similarity for patch token SAE features at threshold 0.001.

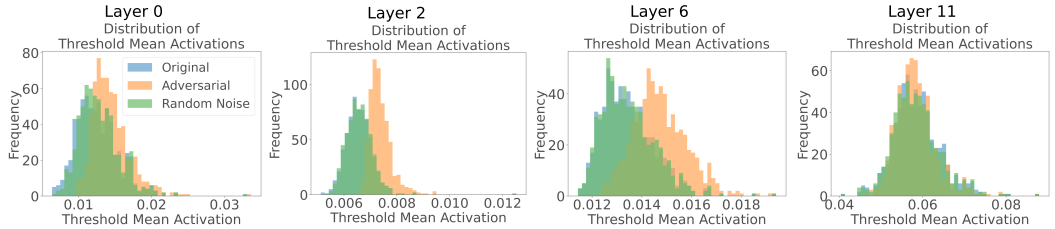


Figure 24: Histogram of mean activations for patch token SAE features at threshold 0.001, with noise.

G ACRONYMS

AEx adversarial example

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

ViT vision transformer